Notes: Hennessy & Whited (JF 2007)

How Costly Is External Financing? Evidence from a Structural Estimation

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Contents

1 What the paper is doing

1.1 Research question

The paper asks: *How large are the "true" costs of external funds*—especially external equity—once we account for dynamic, forward-looking corporate policies (investment, payout, leverage, default)?

1.2 Core contribution

Instead of directly measuring issuance fees from underwriting data, the paper estimates financing frictions *indirectly* by fitting a dynamic structural model to observed moments of corporate behavior using simulated method of moments (SMM). The model endogenizes:

- investment and capital accumulation,
- payout vs equity issuance (with fixed + convex flotation costs),
- borrowing vs corporate saving,
- endogenous default with renegotiation and bankruptcy costs.

1.3 Headline findings

- External equity is costly in a way consistent with *large indirect costs* beyond underwriting fees: estimated issuance costs are fixed and convex (linear–quadratic).
- Small firms face larger financing frictions than large firms (both higher equity issuance costs and higher bankruptcy costs).
- "Constraint proxies" are imperfect: in simulated data, several commonly used indicators can *fall* when true financing frictions rise, because firms endogenously adjust leverage, cash, and payout.

2 Model primitives and timing

2.1 Real technology

Let k be capital. Capital depreciates at rate δ. Profitability is captured by a persistent shock z.

Assumption 1 (Operating profits)**.** *Operating profits are*

$$z\pi(k) = zk^{\alpha}, \qquad \alpha \in (0,1). \tag{1}$$

2.2 Shock process

The shock z follows a Markov process. Empirically the paper models:

$$\ln z_{t+1} = \rho \ln z_t + \varepsilon_{t+1}, \qquad \varepsilon_{t+1} \sim (0, \sigma_{\varepsilon}^2),$$

and discretizes this process on a grid for numerical solution.

2.3 One-period debt and corporate saving

The firm can borrow ($b' > 0$) or save ($b' < 0$). If the firm saves, it earns the pre-tax risk-free rate r. If the firm borrows, it uses a one-period defaultable contract with promised yield $\tilde{r}(k', b', z)$ determined by a competitive lender (zero-profit pricing) given the firm's endogenous default behavior.

Interest income is taxed at rate τ_i at the investor level. The paper discounts payoffs at $1/[1 + r(1 - \tau_i)]$ accordingly.

2.4 Taxes

There are three tax objects: corporate income taxes, investor-level interest taxes, and a reduced-form *convex* distribution tax schedule.

Corporate tax bill. Let χ be an indicator for positive taxable income. Corporate taxes are

$$T_c(k', b', z, z') \equiv \left[\tau_c^+ \chi + \tau_c^- (1 - \chi)\right] \cdot \left[z' \pi(k') - \delta k' - \tilde{r}(k', b', z)\, b'\right], \tag{2}$$

with $0 < \tau_c^- < \tau_c^+ < 1$. Intuition: τ_c^+ applies in positive taxable income states and τ_c^- is the effective rebate rate in negative taxable income states.

Taxation of payouts. Cash distributions $X \geq 0$ face a rising marginal tax rate:

$$T_d(X) \equiv \int_0^X \tau_d(x)\, \mathrm{d}x, \tag{3}$$

$$\tau_d(x) \equiv \bar{\tau}_d\big(1 - e^{-\phi x}\big), \qquad \phi > 0, \;\; \bar{\tau}_d \in (0, 1), \tag{4}$$

so

$$T_d(X) = \bar{\tau}_d \left[X - \frac{1 - e^{-\phi X}}{\phi}\right]. \tag{5}$$

Higher ϕ implies stronger convexity and stronger payout smoothing incentives.

2.5 Costs of external equity (linear–quadratic)

Assumption 2 (Equity flotation cost function)**.** *If the firm raises net equity $x \geq 0$, the flotation cost is*

$$\Lambda(x) = \lambda_0 + \lambda_1 x + \lambda_2 x^2, \qquad \lambda_i \geq 0. \tag{6}$$

Interpretation.

- λ_0 (fixed cost) generates *equity inertia*: the firm avoids small issuances.
- λ_1 controls marginal costs near $x = 0$.
- λ_2 makes marginal costs increasing in issuance size.

2.6 Bankruptcy costs

Assumption 3 (Deadweight bankruptcy costs)**.** *If default occurs, deadweight costs are*

$$BC(k') = \xi(1-\delta)k', \tag{7}$$

so $(1-\xi)(1-\delta)k'$ *is the surviving value of installed capital in default.*

3 State variables, financing identity, and the Bellman equation

3.1 Realized net worth and revised net worth

Given choices (k', b') and next shock z', realized net worth is

$$w(k', b', z, z') \equiv (1-\delta)k' + z'\pi(k') - T_c(k', b', z, z') - (1 + \tilde{r}(k', b', z))b'. \tag{8}$$

The model uses *revised net worth* as the state variable:

$$\tilde{w}(k', b', z, z') \equiv \max\{\bar{w}(z'),\, w(k', b', z, z')\}. \tag{9}$$

The default schedule $\bar{w}(z')$ is determined in equilibrium (Section 4).

3.2 Within-period financing identity (payout vs issuance)

At the decision stage with state $(\tilde{w}, z)$, choosing (k', b') implies:

- payout (if feasible): $d = \tilde{w} + b' - k'$ when $\tilde{w} + b' \geq k'$,
- equity issuance (if needed): $e = k' - \tilde{w} - b'$ when $k' > \tilde{w} + b'$.

Define indicator functions:

$$\Phi^d = \mathbf{1}\{\tilde{w} + b' - k' \geq 0\}, \qquad \Phi^i = \mathbf{1}\{k' - \tilde{w} - b' > 0\}.$$

3.3 Bellman equation

Let $V(\tilde{w}, z)$ denote equity value. The Bellman equation can be written as

$$\begin{aligned} V(\tilde{w}, z) = \max_{(k',b')\in\Gamma(z)} \Bigg\{ & \Phi^d \Big[\tilde{w} + b' - k' - T_d(\tilde{w} + b' - k')\Big] - \Phi^i \Big[k' - \tilde{w} - b' + \Lambda(k' - \tilde{w} - b')\Big] \\ & + \frac{1}{1 + r(1-\tau_i)} \int_{z'_d(k',b',z)}^{\bar{z}} V\big(w(k', b', z, z'), z'\big)\ Q(z,\, \mathrm{d}z') \Bigg\}, \end{aligned} \tag{10}$$

where $z'_d(k', b', z)$ is the default-inducing shock threshold. (Because $V(\bar{w}(z'), z') = 0$, the continuation value integrates only over the no-default region.)

3.4 Basic value-function properties (why default is well behaved)

The paper uses monotonicity and continuity arguments to show:

- $V(\tilde{w}, z)$ is strictly increasing in $\tilde{w}$ (more net worth raises value),
- $V(\tilde{w}, z)$ is nondecreasing in z (better profitability weakly raises value),
- these imply a unique default schedule $\bar{w}(z)$ and a unique default threshold z'_d for each choice (k', b').

4 Default, recovery, and the debt pricing equilibrium

4.1 Default schedule (reservation value = 0)

Equity's value in renegotiation is normalized to zero. Therefore $\bar{w}(z')$ is defined by

$$V(\bar{w}(z'), z') = 0. \tag{11}$$

By strict monotonicity of V in $\tilde{w}$, $\bar{w}(z')$ is unique and (in this model) negative.

4.2 Default threshold

For given (k', b', z), realized net worth $w(k', b', z, z')$ is increasing in z' while $\bar{w}(z')$ is nonincreasing in z'. Thus there exists a unique $z'_d(k', b', z)$ such that

$$w(k', b', z, z'_d) = \bar{w}(z'_d). \tag{12}$$

Default occurs for $z' < z'_d$; repayment occurs for $z' \geq z'_d$.

4.3 Bank recovery in default

In default, the lender extracts value and equity is pushed down to its reservation value $\bar{w}(z')$. The recovery is

$$R(k', z') = (1 - \xi)(1 - \delta)k' + z'\pi(k') - \left[\tau_c^+ \chi + \tau_c^- (1 - \chi)\right] \cdot \left[z'\pi(k') - \delta k'\right] - \bar{w}(z'). \tag{13}$$

Interpretation:

- Bankruptcy costs destroy $\xi(1 - \delta)k'$.
- Interest deductions are disallowed in default in the paper's formulation (hence the tax term excludes interest).
- $-\bar{w}(z')$ represents the going-concern surplus transferred to the lender under renegotiation.

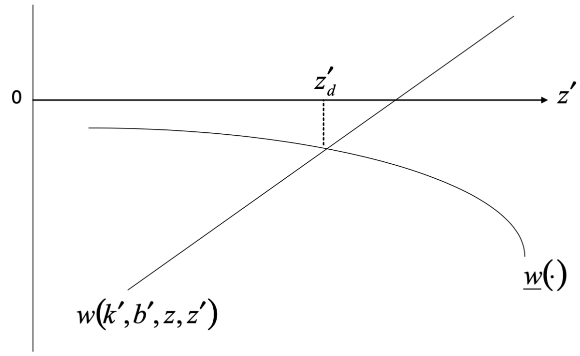

Figure 1. Endogenous default. This figure depicts the determination of the critical shock, z'_d, at which the firm chooses to default. On the horizontal axis is the shock, z'. $w(k',b',z,z')$ is the firm's realized net worth, and $\underline{w}(\cdot)$ is the firm's default schedule.

Figure 1. Endogenous default.

4.4 Competitive debt pricing (zero profit)

Debt is priced so that the expected discounted lender payoff equals b'. In reduced form:

$$b' = \frac{1}{1 + r(1-\tau_i)} \left([1 + (1-\tau_i)\tilde{r}]b' \cdot \Pr(z' \geq z'_d) + \int_{z'<z'_d} R(k', z')\, Q(z, \mathrm{d}z') \right). \tag{14}$$

This pins down $\tilde{r} = \tilde{r}(k', b', z)$ given the policy-induced default region.

4.5 Figure 1

Figure 1 visualizes endogenous default: plot $w(k', b', z, z')$ (increasing in z') and $\bar{w}(z')$ (nonincreasing) and take the intersection as the default cutoff z'_d.

5 Optimality conditions and mechanisms

The paper derives first-order conditions for financing and investment. I summarize them conceptually (the exact algebra is in the paper).

5.1 Financial optimality (equity vs debt at the margin)

At an interior optimal financing policy (issuing equity or paying dividends at the margin), the condition is:

$$\text{marginal cost of equity finance} = \text{discounted shadow cost of servicing one more dollar of debt.}$$

Special case: if there are no distribution taxes and no equity costs, the condition collapses to a trade-off theory statement:

$$\text{marginal tax shield benefit} = \text{marginal expected bankruptcy cost.}$$

5.2 Investment optimality and the agency cost of debt

Investment (choice of k') trades off current marginal cost against discounted expected marginal value of installed capital. A key term appears in the marginal value:

$$-b' \frac{\partial \tilde{r}}{\partial k'}.$$

Increasing k' reduces default risk and lowers the borrowing cost schedule, benefiting debtholders. Because debt is priced competitively as a function of (k', b'), equity internalizes part of this benefit when choosing k'; nevertheless, limited liability still creates an underinvestment incentive relative to the first best.

5.3 Equity inertia region

Because $\lambda_0 > 0$, small issuances can be dominated by fixed costs. Then the firm may neither pay dividends nor issue equity (an "inertia region"), and debt becomes the marginal financing source for small investment adjustments. The paper derives an adjusted investment condition for this region.

6 Numerical solution, simulation, and SMM estimation

6.1 Solving the dynamic program

No closed form exists. The solution iterates:

1. guess the debt yield schedule $\tilde{r}(k', b', z)$,
2. solve the Bellman equation for $V(\tilde{w}, z)$ and policies,
3. update $\tilde{r}$ using the implied default set and the lender pricing condition,
4. iterate until convergence.

6.2 Simulated panels

To mimic Compustat, the paper simulates 20,000 firms for 200 periods and retains the last 14 periods. It repeats this for $S = 6$ artificial panels.

6.3 Estimated parameters

The estimated parameter vector is

$$\theta = (\alpha, \lambda_0, \lambda_1, \lambda_2, \xi, \phi, \sigma_\varepsilon, \rho).$$

Tax rates and depreciation are calibrated from external evidence.

6.4 Moment selection (identification logic)

Moments are chosen because they are informative about the parameters:

- Equity issuance mean/variance/frequency and the covariance of issuance with investment identify $(\lambda_0, \lambda_1, \lambda_2)$.
- Mean leverage and the covariance between investment and leverage identify bankruptcy costs ξ (and debt's position in the pecking order).
- Payout ratio and payout variance identify the payout tax curvature ϕ.
- Investment variance identifies α (profit curvature).
- Income serial correlation and innovation volatility identify $(\rho, \sigma_\varepsilon)$.

6.5 SMM objective function (Appendix B)

Let $\hat{M}_N$ be data moments and $\hat{m}_n^{(s)}(\theta)$ simulated moments (simulation s). The SMM estimator solves

$$\hat{\theta} = \arg\min_{\theta} \left[\hat{M}_N - \frac{1}{S}\sum_{s=1}^{S} \hat{m}_n^{(s)}(\theta)\right]' \hat{W}_N \left[\hat{M}_N - \frac{1}{S}\sum_{s=1}^{S} \hat{m}_n^{(s)}(\theta)\right], \tag{15}$$

with optimal $\hat{W}_N \approx [N\mathrm{Var}(\hat{M}_N)]^{-1}$. The minimized distance yields an overidentification test statistic that is asymptotically χ^2.

7 How to interpret the tables and figure

7.1 Figure 1: endogenous default

What it is. A picture of the default boundary: default happens when realized net worth falls below the endogenous schedule $\bar{w}(z')$.
How to read it. More leverage (b' up) or less planned capital (k' down) shifts the realized net worth curve down, raising default probability and increasing the borrowing rate schedule.

7.2 Table I: full-sample SMM fit

Panel A (moments). Compares empirical moments to simulated moments. Good fit here means the model reproduces:

- equity issuance behavior (level, dispersion, and how often it happens),
- payout behavior (payout ratio, payout volatility),
- corporate saving (frequency of negative debt),
- leverage and its comovement with investment,
- shock dynamics of operating income.

Panel A: Moments		
	Actual Moments	Simulated Moments
Average Equity Issuance/Assets	0.0892	0.0963
Variance of Equity Issuance/Assets	0.0911	0.0847
Variance of Investment/Assets	0.0068	0.0117
Frequency of Equity Issuance	0.1751	0.2305
Payout Ratio	0.2226	0.2026
Frequency of Negative Debt	0.3189	0.3258
Variance of Distributions	0.0013	0.0037
Average Debt-Assets Ratio (Net of Cash)	0.1204	0.1104
Covariance of Investment and Equity Issuance	0.0004	0.0005
Covariance of Investment and Leverage	−0.0018	−0.0025
Serial Correlation of Income/Assets	0.5121	0.5661
Standard Deviation of the Shock to Income/Assets	0.1185	0.1057

Panel B: Parameter Estimates

α	λ_0	λ_1	λ_2	ξ	ϕ	σ_ε	ρ	χ^2
0.627	0.598	0.091	0.0004	0.104	0.732	0.118	0.684	8.018
(0.219)	(0.233)	(0.026)	(0.0008)	(0.059)	(0.844)	(0.042)	(0.349)	(0.091)

Table I Simulated Moments Estimation for the Full Sample

Panel B (parameters). Reads as:

- ξ is economically meaningful: bankruptcy costs are about ξ times depreciated capital.
- λ_0 implies lumpiness of equity issuance.
- λ_1 and λ_2 shape marginal issuance costs.
- ϕ implies payout smoothing incentives via convex payout taxation.
- $(\rho, \sigma_\varepsilon)$ recover the persistence/volatility of the profitability process.

7.2.1 Mapping flotation costs to underwriting fee schedules (equations (33)–(34) in the paper)

To compare to underwriting data, define gross proceeds p and underwriting fee c. Net proceeds are $p - c$. The model implies

$$c = \lambda_0 + \lambda_1(p - c) + \lambda_2(p - c)^2. \tag{16}$$

Implicit differentiation yields

$$\frac{\partial c}{\partial p} = 1 - \frac{1}{1 + \lambda_1 + 2\lambda_2(p - c)}, \tag{17}$$

$$\frac{\partial^2 c}{\partial p^2} = 2\lambda_2 \left[1 + \lambda_1 + 2\lambda_2(p - c)\right]^{-3}. \tag{18}$$

Hence at the origin ($p \to 0$ so $p - c \to 0$), the initial marginal underwriting fee share is approximately

$$\left.\frac{\partial c}{\partial p}\right|_{p \to 0} \approx \frac{\lambda_1}{1 + \lambda_1}. \tag{19}$$

7.3 Tables II and III: small vs large firms

Panel A: Moments

	Actual Moments	Simulated Moments
Average Equity Issuance/Assets	0.1243	0.1389
Variance of Equity Issuance/Assets	0.1400	0.1352
Variance of Investment/Assets	0.0081	0.0171
Frequency of Equity Issuance	0.2391	0.3307
Payout Ratio	0.1252	0.0971
Frequency of Negative Debt	0.4284	0.3990
Variance of Distributions	0.0031	0.0052
Average Debt-Assets Ratio (Net of Cash)	0.0840	0.0614
Covariance of Investment and Equity Issuance	0.0007	0.0010
Covariance of Investment and Leverage	−0.0011	−0.0019
Serial Correlation of Income/Assets	0.4083	0.4123
Standard Deviation of the Shock to Income/Assets	0.1662	0.1419

Panel B: Parameter Estimates

α	λ_0	λ_1	λ_2	ξ	ϕ	σ_ε	ρ	χ^2
0.693	0.951	0.120	0.0004	0.151	0.831	0.159	0.498	9.182
(0.302)	(0.495)	(0.039)	(0.0010)	(0.072)	(0.711)	(0.081)	(0.280)	(0.057)

Table II Simulated Moments Estimation for Small Firms

Panel A: Moments

	Actual Moments	Simulated Moments
Average Equity Issuance/Assets	0.0662	0.0765
Variance of Equity Issuance/Assets	0.0507	0.0784
Variance of Investment/Assets	0.0052	0.0126
Frequency of Equity Issuance	0.1470	0.1905
Payout Ratio	0.3212	0.2852
Frequency of Negative Debt	0.2282	0.2402
Variance of Distributions	0.0009	0.0013
Average Debt-Assets Ratio (Net of Cash)	0.1452	0.1610
Covariance of Investment and Equity Issuance	0.0004	0.0009
Covariance of Investment and Leverage	−0.0015	−0.0031
Serial Correlation of Income/Assets	0.6455	0.6340
Standard Deviation of the Shock to Income/Assets	0.0796	0.0735

Panel B: Parameter Estimates

α	λ_0	λ_1	λ_2	ξ	ϕ	σ_ε	ρ	χ^2
0.577	0.389	0.053	0.0002	0.084	0.695	0.086	0.791	7.145
(0.235)	(0.302)	(0.022)	(0.0003)	(0.055)	(0.920)	(0.045)	(0.322)	(0.128)

Table III Simulated Moments Estimation for Large Firms

What changes. The paper re-estimates θ in subsamples defined by firm size (dropping the middle third).

Main message. Small firms face larger financing frictions:

- higher bankruptcy cost parameter ξ,
- larger external equity costs $(\lambda_0, \lambda_1, \lambda_2)$,
- more volatile and less persistent shocks (higher σ_ε, lower ρ).

Why the shock process matters. Even if large firms have smaller issuance costs, their more stable environments (high ρ, low σ_ε) can reduce the need to issue equity often.

Interpreting the paper's "marginal flotation costs start at ..." numbers. Using $\partial c/\partial p \approx \lambda_1/(1+\lambda_1)$ at the origin:

- large firms: $\lambda_1 \approx 0.053 \Rightarrow 5.0\%$,
- small firms: $\lambda_1 \approx 0.120 \Rightarrow 10.7\%$.

7.4 Table IV: constrained vs unconstrained firms in simulated data

Experiment. "Unconstrained" sets financing frictions to zero; "constrained" uses estimated frictions.

Reading. Financing frictions lower investment and firm value (Tobin's q), and change leverage/payout/issuance patterns. Effects are stronger for the small-firm calibration.

7.5 Table V: debt underwriting fees robustness

What it is. Re-estimation under an added proportional fee on debt issuance (calibrated from underwriting evidence).

	Investment/ Assets	Tobin's q	Net Debt/ Assets	Equity Issuance/ Assets	Frequency of Equity Issuance
Large Firms					
Constrained	0.140	1.592	0.160	0.077	0.191
Unconstrained	0.151	1.913	0.178	0.081	0.203
Small Firms					
Constrained	0.125	2.913	0.061	0.139	0.331
Unconstrained	0.159	3.724	0.099	0.142	0.386

Table IV Summary Statistics from Simulated Constrained and Unconstrained Firms

	α	λ_0	λ_1	λ_2	ξ	ϕ	σ_ε	ρ	χ^2
Full Sample									
	0.643	0.601	0.095	0.0004	0.109	0.780	0.122	0.669	7.665
	(0.288)	(0.259)	(0.036)	(0.0003)	(0.063)	(0.752)	(0.040)	(0.351)	(0.105)
Large Firms									
	0.589	0.411	0.061	0.0002	0.073	0.837	0.091	0.732	7.146
	(0.249)	(0.283)	(0.020)	(0.0006)	(0.058)	(1.003)	(0.051)	(0.328)	(0.282)
Small Firms									
	0.701	1.010	0.129	0.0005	0.141	0.702	0.152	0.511	9.343
	(0.327)	(0.481)	(0.037)	(0.0009)	(0.066)	(0.824)	(0.073)	(0.271)	(0.025)

Table V Estimates of the Cost of External Funds: Debt Issuance Costs

Takeaway. Parameter estimates remain similar, so the baseline inference about large equity issuance costs and nontrivial bankruptcy costs is not sensitive to assuming zero debt underwriting fees.

7.6 Table VI: alternative sample splits (dividends; constraint indexes)

What it is. Re-estimation in subsamples split by: dividend payout, the Whited–Wu (WW) index, the Cleary index, and the Kaplan–Zingales (KZ) index (dropping the middle third).
How to read. A split is a good *classifier of cost-of-funds* if the "high constraint" group has higher estimated $(\lambda_0, \lambda_1, \lambda_2, \xi)$.

Main conclusions from the table.

- **Low dividends:** higher equity issuance costs and higher bankruptcy costs than high-dividend firms.
- **High WW index and high Cleary index:** systematically higher estimated equity issuance costs and bankruptcy costs, consistent with these indexes capturing high costs of external funds.
- **KZ index:** split is less clean for equity issuance costs (some equity-cost parameters can be higher in the "low KZ" group), suggesting KZ may capture a different object (e.g., equilibrium financing/payout mix) rather than a monotone cost-of-funds ranking.

	α	λ_0	λ_1	λ_2	ξ	ϕ	σ_ε	ρ	χ^2
Low Dividends	0.671	0.624	0.097	0.0004	0.117	0.821	0.135	0.590	8.599
	(0.303)	(0.362)	(0.025)	(0.0003)	(0.062)	(0.680)	(0.051)	(0.298)	(0.072)
High Dividends	0.588	0.484	0.074	0.0003	0.089	0.683	0.104	0.721	6.548
	(0.234)	(0.374)	(0.039)	(0.0007)	(0.066)	(0.754)	(0.037)	(0.366)	(0.162)
High WW Index	0.688	1.002	0.123	0.0005	0.140	0.990	0.138	0.483	8.322
	(0.327)	(0.481)	(0.037)	(0.0003)	(0.050)	(0.951)	(0.033)	(0.267)	(0.080)
Low WW Index	0.563	0.494	0.062	0.0003	0.083	0.633	0.097	0.704	6.889
	(0.200)	(0.189)	(0.018)	(0.0010)	(0.063)	(0.864)	(0.039)	(0.347)	(0.142)
High Cleary Index	0.702	0.808	0.108	0.0004	0.133	0.769	0.124	0.605	10.147
	(0.395)	(0.367)	(0.045)	(0.0008)	(0.042)	(0.882)	(0.052)	(0.482)	(0.038)
Low Cleary Index	0.644	0.466	0.063	0.0003	0.072	0.571	0.102	0.532	6.504
	(0.362)	(0.301)	(0.032)	(0.0007)	(0.064)	(0.914)	(0.054)	(0.269)	(0.164)
High KZ Index	0.611	0.502	0.082	0.0003	0.110	0.887	0.127	0.655	6.941
	(0.377)	(0.270)	(0.038)	(0.0006)	(0.055)	(1.079)	(0.056)	(0.307)	(0.139)
Low KZ Index	0.603	0.655	0.107	0.0004	0.073	0.416	0.103	0.692	7.628
	(0.280)	(0.359)	(0.039)	(0.0007)	(0.059)	(0.652)	(0.049)	(0.315)	(0.106)

Table VI Estimates of the Cost of External Funds: Alternative Sample Splits

	Baseline Moments	α	λ_0	λ_1	λ_2	ξ	ϕ	σ_e	ρ
Average Equity Issuance/Assets	0.0892	0.3976	−1.1545	−0.7550	−0.1010	1.3606	0.0214	1.7625	−0.9539
Variance of Equity Issuance/Assets	0.0911	0.6034	−0.6356	−0.6303	−0.0040	0.5033	0.0262	1.3983	−0.6883
Variance of Investment/Assets	0.0068	1.0371	−0.4725	−0.1962	−0.4607	0.4079	−0.4539	1.3979	0.5361
Frequency of Equity Issuance	0.1751	0.7305	−0.8172	−0.7319	−0.1561	1.3937	0.1256	1.0418	−0.4435
Payout Ratio	0.2226	0.0334	0.0407	0.0170	0.0184	−0.1778	−0.8514	−0.6841	0.2986
Frequency of Negative Debt	0.3189	0.3408	−0.2098	−0.2567	0.0900	0.5328	0.5967	0.7004	0.2139
Variance of Distributions	0.0013	0.4529	−0.0761	0.1819	0.1975	0.0069	−0.5561	1.6851	0.2931
Average Net Debt−Assets Ratio	0.1204	−0.3109	0.5373	0.1052	−0.0235	−1.2708	−0.6710	−0.8551	0.2538
Covariance of Investment and Equity Issuance	0.0004	0.4821	−0.3826	−0.8728	−0.0025	0.8456	−0.0674	0.8571	−0.0561
Covariance of Investment and Leverage	−0.0018	0.5002	0.8213	0.4954	0.1626	−1.6183	−0.7967	0.5636	0.1923
Serial Correlation of Income/Assets	0.5121	−0.1065	−0.0913	0.0696	0.0713	0.6314	0.0870	−0.0598	1.5608
Standard Deviation of the Shock to Incomes/Assets	0.1185	−0.6122	−0.0686	0.0553	0.0184	0.8591	0.0822	1.0443	−0.5471

Table VII Sensitivity of Model Moments to Parameters

7.7 Tables VII and VIII: elasticities as "mechanism diagnostics"

7.7.1 Elasticity calculation (equation (35) in the paper)

For an outcome y and parameter κ with baseline estimate $\hat{\kappa}$, define $\underline{\kappa} = 0.5\hat{\kappa}$ and $\overline{\kappa} = 1.5\hat{\kappa}$. The elasticity reported is

$$\epsilon_{y,\kappa} \equiv \frac{y(\overline{\kappa}) - y(\underline{\kappa})}{\overline{\kappa} - \underline{\kappa}} \cdot \frac{\hat{\kappa}}{y(\hat{\kappa})}. \tag{20}$$

Interpretation: a local (finite-difference) elasticity around the baseline calibration.

7.7.2 Table VII: sensitivity of model moments to parameters

What the rows are. Moments like mean issuance, issuance variance, investment variance, leverage, comovements, and shock moments.

What the columns are. Elasticities w.r.t. α, the equity cost parameters $\lambda_0, \lambda_1, \lambda_2$, bankruptcy costs ξ, payout tax curvature ϕ, and shock process parameters $(\sigma_\varepsilon, \rho)$.

Reading the key sign patterns.

	Baseline Moments	α	λ_0	λ_1	λ_2	ξ	ϕ	σ_e	ρ
Investment-q Sensitivity	0.0061	0.0475	−0.3782	−0.4214	−0.1757	−0.0187	0.1428	0.4009	0.3967
Investment–Cash Flow Sensitivity	0.1917	−1.0023	−0.5493	−0.5089	−0.1577	0.5584	0.0016	0.1174	0.5093
WW Index	0.6741	0.0108	0.2890	0.2785	0.0304	−0.0390	−0.2482	−0.1746	−0.2884
Cleary Index	−0.2422	−0.1772	0.8579	0.7516	0.0211	−0.2821	−0.2499	−0.6485	0.3338
KZ Index	0.9010	0.5685	−0.0150	−0.5388	−0.0899	−0.3464	−0.5241	−0.3504	0.4543

Table VIII: Sensitivity of Finance constraint Indicators to Parameters

- *Equity issuance moments:* increasing any λ reduces mean/frequency of equity issuance (negative elasticities), consistent with equity moving down the pecking order.
- *Investment variance:* increasing equity costs reduces investment volatility (the firm responds less aggressively to profitability changes when external equity is expensive).
- *Leverage and saving:* increasing bankruptcy costs ξ lowers average leverage and increases negative debt frequency (more precautionary saving / less borrowing).
- *Comovements:* when equity becomes more costly, $\mathrm{Cov}(I, \text{equity issuance})$ falls and $\mathrm{Cov}(I, \text{leverage})$ rises; when bankruptcy costs rise, the opposite occurs (debt becomes less attractive).
- *Payout moments:* increasing ϕ (more convex payout taxation) reduces payout ratio and payout variance (stronger smoothing incentives).
- *Shock moments:* increasing ρ raises income persistence; increasing σ_ε increases shock volatility, which in turn pushes firms toward more precautionary behavior.

7.7.3 Table VIII: sensitivity of common financing-constraint indicators

Constraint indicators studied.

1. **Investment–q sensitivity** and **investment–cash flow sensitivity**: coefficients from the regression
$$\frac{I_{it}}{K_{it}} = a + b_q q_{it} + b_{cf} \frac{CF_{it}}{K_{it}} + u_{it}. \tag{21}$$
2. **KZ index, WW index, Cleary index**: mechanical functions of observables (defined in Appendix C; see Section 8 below).

Key messages from the elasticities.

- In the simulated model, *investment–q sensitivity declines* as financing frictions rise (consistent with frictions dampening responsiveness to marginal q).
- *Investment–cash flow sensitivity* declines with higher equity issuance costs (because the firm responds less aggressively to shocks when equity is expensive), but can increase with higher bankruptcy costs (endogenous leverage reductions change how cash flow maps into investment).
- Indexes (KZ/WW/Cleary) can move in non-monotone ways because leverage, payout, and cash holdings are *endogenous outcomes* that respond to frictions.

Interpretation for empirical work. A proxy is not "wrong" because it is not monotone in a parameter—rather, the table shows that proxies may measure *equilibrium financial choices* rather than the underlying structural cost parameters. This is the paper's cautionary contribution.

8 Constraint indexes used in the paper (measurement models)

The paper uses three popular finance-constraint indexes and computes them in both data and simulated data. This is important because Tables VI and VIII interpret these indexes as *classifiers* or *proxies* for costly external finance.

8.1 Kaplan–Zingales (KZ) index

The paper uses a "synthetic" KZ index:

$$\text{KZ} = -1.001909\,CF + 3.139193\,TLTD - 39.36780\,TDIV - 1.314759\,CASH + 0.2826389\,Q, \tag{22}$$

where: CF is cash flow/book assets, $TLTD$ is total long-term debt/book assets, $TDIV$ is total dividends/book assets, $CASH$ is cash stock/book assets, and Q is market-to-book.

8.2 Whited–Wu (WW) index

$$\text{WW} = -0.091\,CF - 0.062\,DIVPOS + 0.021\,TLTD - 0.044\,LNTA + 0.102\,ISG - 0.035\,SG, \tag{23}$$

where $DIVPOS$ is an indicator for paying dividends, $LNTA$ is log(book assets), SG is own-firm real sales growth, and ISG is industry sales growth.

8.3 Cleary index

The Cleary index is constructed from discriminant analysis and (as written in the paper) is:

$$\begin{aligned}\text{Cleary} = -0.11905\,CURAT - 1.903670\,TLTD + 0.00138\,COVER \\ + 1.45618\,IMARG + 2.03604\,SG - 0.04772\,SLACK.\end{aligned} \tag{24}$$

The paper multiplies this index by -1 so that it increases with the likelihood of facing costly external finance (to align its direction with KZ and WW).

9 Summmary

1. Structural objects: $(\lambda_0, \lambda_1, \lambda_2)$ for equity issuance costs; ξ for bankruptcy costs.
2. Mapping to underwriting fee shares: $\partial c/\partial p|_{p\to 0} \approx \lambda_1/(1+\lambda_1)$.
3. Mechanism insight: endogenous leverage/cash/payout mean "constraint proxies" can be non-monotone in true frictions.
4. Method: SMM—choose parameters so simulated moments match data moments.